

Interpretable Gene Programs or Foundation Models?

Comparing Sparse Dictionary Learning and scGPT for Single-Cell Classification

Motivation and Research Question

Single-cell foundation models such as scGPT learn representations from large collections of gene-expression data and can transfer these representations to downstream biological tasks. However, their learned embedding dimensions are difficult to interpret directly. This raises a natural machine-learning question: what predictive value does a pretrained foundation-model representation provide beyond a simpler representation whose latent structure can be inspected?

We propose a controlled comparison between **pretrained scGPT** and **sparse dictionary learning** for single-cell representation learning. Both methods will transform the same single-cell observations into learned representations, which will then be evaluated using the same downstream classifier.

Our central research question is:

What predictive value does a pretrained scGPT representation provide beyond an interpretable sparse representation learned directly from the target single-cell data?

The project studies two complementary aspects of representation quality:

1. **Predictive utility:** How useful is each representation for downstream cell-type classification?
2. **Interpretability:** What biological structure can be identified directly from the learned representation?

Rather than treating downstream accuracy as the only criterion, we aim to characterize the tradeoff between predictive performance and interpretability in single-cell representation learning.

Dataset and Preprocessing

We will use a public single-cell RNA-sequencing dataset with gene-expression measurements and biological cell-type annotations, such as the Pancreas benchmark available through `scvi-tools`.

Let $X \in \mathbb{R}^{N \times p}$ denote the processed cell-by-gene expression matrix, where N is the number of cells and p is the number of retained genes.

We will construct fixed training and test partitions before fitting data-dependent components of the classical representation pipeline. Where meaningful donor, sample, or sequencing-platform metadata are available, we will consider grouped splitting to reduce dependence between training and evaluation observations.

Highly variable gene selection and other data-dependent preprocessing for dictionary learning will be fit using training observations only. The preprocessing and gene sets used by scGPT will follow the requirements of the pretrained model rather than forcing both methods to use an artificial common input representation.

Sparse Dictionary Learning

Dictionary learning approximates the expression matrix using a collection of latent basis vectors, or dictionary atoms. We model

$$X \approx ZD,$$

where $D \in \mathbb{R}^{K \times p}$ contains K dictionary atoms and $Z \in \mathbb{R}^{N \times K}$ contains the representation of each cell in terms of those atoms.

We will estimate the representation using a sparse dictionary-learning objective of the form

$$\min_{Z, D} \frac{1}{2} \|X - ZD\|_F^2 + \lambda \|Z\|_1, \quad \|D_k\|_2 \leq 1.$$

The reconstruction term encourages the dictionary to capture recurring expression structure, while the L_1 penalty encourages each cell to use only a small number of dictionary atoms.

Each row z_i of Z therefore provides a low-dimensional representation of cell i . The corresponding atoms in D can be inspected through their gene weights, allowing us to identify which genes characterize each learned expression program.

The dictionary size K and sparsity parameter λ will be selected using a modest prespecified training/validation procedure. We will avoid large hyperparameter searches so that the project remains focused on the representation comparison rather than extensive model tuning.

Pretrained scGPT Representation

For the foundation-model representation, each cell will be passed through a pretrained scGPT model to obtain a fixed embedding,

$$x_i \longrightarrow z_i^{\text{scGPT}}.$$

The pretrained model will remain frozen; we will not fine-tune scGPT for the downstream classification task. This allows the experiment to evaluate the information already contained in its pretrained representation rather than introducing additional supervised representation learning.

Once generated, scGPT embeddings will be cached and treated as fixed features for all downstream analyses.

Thus, the two primary representation pipelines are

$$X \longrightarrow Z_{\text{DL}} \longrightarrow \text{classifier},$$

and

$$X \longrightarrow Z_{\text{scGPT}} \longrightarrow \text{classifier}.$$

A classifier using the processed gene-expression features directly will provide an additional reference baseline.

Controlled Downstream Comparison

To isolate differences in representation quality, both learned representations will feed into the same downstream model family: L_2 -regularized multinomial logistic regression.

For representation z_i , the classifier takes the form

$$P(Y = c \mid z_i) = \frac{\exp(\beta_c^\top z_i)}{\sum_{k=1}^C \exp(\beta_k^\top z_i)}.$$

Regularization will be tuned using the same training-only procedure and search budget for both representations.

The primary comparison will therefore be

$$\begin{array}{llll} X & \rightarrow & \text{raw expression} & \rightarrow \text{logistic regression,} \\ X & \rightarrow & \text{sparse dictionary codes} & \rightarrow \text{logistic regression,} \\ X & \rightarrow & \text{pretrained scGPT embeddings} & \rightarrow \text{logistic regression.} \end{array}$$

Holding the downstream model fixed makes the learned representation the principal difference between the two representation-learning pipelines.

Evaluation

Macro-F1 will be the primary classification metric because cell-type frequencies may be unequal. We will additionally report balanced accuracy, per-cell-type recall, and multiclass log loss.

The main predictive comparison asks whether pretrained scGPT embeddings provide additional linearly accessible information about cell type relative to sparse dictionary codes learned from the target dataset.

We will report performance differences on the same held-out cells rather than treating model results as independent experiments.

Our primary predictive figure will compare held-out classification performance across raw expression, dictionary learning, and pretrained scGPT representations.

Representation Interpretability

Predictive performance alone does not describe what each representation has learned. Dictionary learning provides an explicit connection between latent representation dimensions and genes through the dictionary matrix D .

For each dictionary atom D_k , we will examine genes with the largest absolute weights and identify the cell types for which the corresponding activation z_{ik} is strongest. This allows us to characterize atoms as recurring expression programs and determine whether particular programs are associated with specific cell populations.

Where appropriate, selected dictionary atoms may also be compared with established gene sets or biological pathways. Such analyses will be treated as interpretation of the learned representation rather than evidence that every dictionary atom corresponds to a unique biological mechanism.

For scGPT, we will examine how cells organize in the pretrained embedding space and how strongly cell-type information can be recovered by the common linear classifier. We will not claim that individual scGPT embedding coordinates have the same direct interpretation as dictionary atoms.

The comparison therefore intentionally contrasts two different representation-learning philosophies:

$$\underbrace{\text{sparse, directly inspectable latent factors}}_{\text{dictionary learning}} \quad \text{vs.} \quad \underbrace{\text{large-scale pretrained nonlinear representation}}_{\text{scGPT}}.$$

Theoretical Component

The theoretical portion of the project will focus on sparse representation learning. The objective

$$\min_{Z,D} \frac{1}{2} \|X - ZD\|_F^2 + \lambda \|Z\|_1$$

illustrates a fundamental tradeoff between representation fidelity and sparsity.

For fixed D , estimating a cell representation reduces to a sparse-coding problem,

$$\hat{z}_i = \arg \min_z \left\{ \frac{1}{2} \|x_i - zD\|_2^2 + \lambda \|z\|_1 \right\},$$

connecting dictionary learning directly to L_1 -regularized estimation.

As λ increases, representations become increasingly sparse, reducing the number of active latent factors used to describe each cell. Smaller λ allows more atoms to participate, potentially improving reconstruction while reducing sparsity and interpretability.

Joint estimation of Z and D is nonconvex, but each block can be optimized conditionally while holding the other fixed. We will discuss this optimization structure and the role of the dictionary-size and sparsity constraints in controlling representation capacity.

This theoretical analysis provides a classical statistical-learning perspective on representation learning that can be contrasted empirically with the pretrained representation produced by scGPT.

Expected Contribution

The goal is not simply to determine whether scGPT achieves higher classification accuracy than dictionary learning. Instead, the project asks what is gained by moving from an interpretable representation learned from the target dataset to a large pretrained foundation-model representation.

If scGPT substantially improves downstream classification, the results would provide evidence that its pretrained representation contains useful cell-type information not captured by the sparse dictionary representation. If performance is similar, the results would show that a substantially simpler and directly inspectable representation can recover much of the information needed for the evaluated task.

Dictionary learning additionally provides explicit gene-level structure that is unavailable from scGPT embeddings in the same form. The project can therefore characterize both predictive differences and the interpretability obtained from the classical representation.

The broader contribution is a controlled comparison of two approaches to representation learning on the same scientific problem: one based on sparse, interpretable structure learned directly from the target data and one based on large-scale pretrained deep learning.

Computational and Reproducibility Plan

The project is designed to remain computationally manageable. Pretrained scGPT embeddings will be extracted once and cached. Dictionary learning will operate on a reduced set of highly variable genes and can be fit using CPU-based mini-batch algorithms.

All data-dependent classical preprocessing and dictionary estimation will use training observations only. Hyperparameter selection will be performed without access to the final test outcomes.

We will use fixed data partitions, random seeds, cached representations, and scripted evaluation so that the complete comparison can be reproduced.

Timeline and Responsibilities

Weeks	Milestone
1–2	Data preprocessing, splits, and scGPT embeddings
3–4	Dictionary-learning implementation and tuning
5	Downstream classification experiments
6	Dictionary-atom and gene-program interpretation
7	Theoretical analysis, robustness checks, and figures
8	Reproducible code, final report, and presentation

For a four-person group, responsibilities can be divided among data/preprocessing, scGPT representations, dictionary learning and theory, and downstream evaluation/interpretability. All members will contribute to verification, interpretation, writing, and presentation.

References

1. Cui, H. et al. *scGPT: Toward Building a Foundation Model for Single-Cell Multi-omics Using Generative AI*. Nature Methods, 2024.
2. DenAdel et al. *Evaluating the role of pretraining dataset size and diversity on single-cell foundation model performance*. Nature Methods, 2026.
3. Hastie, T., Tibshirani, R., and Friedman, J. *The Elements of Statistical Learning*. Springer.